

CHANGWON, Korea, Republic of

WMO: 471550

Lat: 35.1703N

Lon: 128.5728E

Elev: 38

StdP: 100.86

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.4	(c) -3.6	(d) -21.5	(e) 0.5	(f) -1.2	(g) -19.1	(h) 0.7	(i) -0.3	(j) 6.5	(k) 1.3	(l) 6.0	(m) 0.7	(n) 2.8	(o) 20	(p) 0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.8	(c) 32.6	(d) 25.9	(e) 31.4	(f) 25.4	(g) 30.1	(h) 24.8	(i) 27.1	(j) 30.7	(k) 26.3	(l) 29.8	(m) 25.6	(n) 29.0	(o) 3.2	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	29.3	25.2	20.5	28.5	24.6	19.7	28.0	85.8	30.7	82.2	29.9	79.3	29.2	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.3	(b) 5.4	(c) 4.9	(e) -8.2	(f) 34.5	(g) 2.1	(h) 1.7	(i) -9.7	(j) 35.7	(k) -10.9	(l) 36.7	(m) -12.1	(n) 37.6	(o) -13.6	(p) 38.8		
(a) 6.3	(b) 5.4	(c) 4.9	(e) -10.3	(f) 27.6	(g) 1.7	(h) 1.4	(i) -11.5	(j) 28.6	(k) -12.6	(l) 29.4	(m) -13.5	(n) 30.2	(o) -14.8	(p) 31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	2.8	4.8	8.9	14.1	18.8	22.0	25.7	26.9	22.9	17.8	11.2	4.8
	DBStd	8.65	3.21	3.43	3.16	2.74	2.33	1.84	2.57	2.21	2.35	2.75	3.57	3.50
	HDD10.0	624	224	148	58	3	0	0	0	0	0	1	28	162
	HDD18.3	1979	482	378	292	129	22	1	0	0	1	42	214	418
	CDD10.0	2496	1	2	24	126	273	361	487	525	388	241	65	2
	CDD18.3	809	0	0	0	2	37	111	229	267	138	25	1	0
	CDH23.3	5844	0	0	0	8	160	408	1902	2609	708	49	0	0
	CDH26.7	1658	0	0	0	1	19	52	578	880	126	2	0	0
Wind	WSAvg	2.0	2.1	2.1	2.2	2.2	2.1	2.0	2.0	1.9	1.8	1.7	1.8	2.0
Precipitation	PrecAvg	1585	30	51	86	145	164	194	309	288	167	67	53	30
	PrecMax	2787	69	123	173	286	340	389	766	771	499	213	212	128
	PrecMin	885	0	0	7	30	44	52	75	60	20	0	0	0
	PrecStd	403	20	33	44	68	73	102	176	196	131	54	46	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.8	19.4	24.4	28.7	29.7	34.4	34.6	31.2	26.1	21.4	15.2
		MCWB	9.0	11.4	11.1	15.0	17.5	21.2	26.6	26.6	24.0	18.9	15.5	10.7
	2%	DB	10.5	13.5	17.2	22.2	26.6	28.1	32.6	32.9	29.3	24.7	19.5	13.2
		MCWB	5.8	8.4	11.0	14.0	16.7	20.8	26.3	25.9	22.9	18.1	14.4	8.4
	5%	DB	8.9	11.8	15.6	20.5	25.1	26.9	31.2	31.8	28.1	23.4	18.0	11.6
		MCWB	4.2	6.7	10.1	13.3	16.4	20.4	25.7	25.5	22.0	17.7	13.3	6.8
	10%	DB	7.5	10.2	14.1	19.0	23.6	25.6	29.8	30.7	26.9	22.2	16.5	10.2
		MCWB	3.4	5.4	9.2	12.8	16.1	20.2	25.1	25.2	21.6	16.6	11.9	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	13.5	14.7	18.6	21.1	24.2	28.6	28.2	25.6	22.1	17.7	11.7
		MCDB	12.5	15.1	16.5	20.7	23.3	26.8	32.5	32.2	28.9	24.2	19.8	13.9
	2%	WB	6.9	10.1	13.1	16.7	19.8	23.1	27.2	27.2	24.7	20.5	15.6	9.2
		MCDB	9.2	12.3	15.4	19.4	23.1	25.3	30.7	30.9	27.8	23.1	18.2	12.3
	5%	WB	5.3	7.4	11.2	15.4	18.7	22.4	26.2	26.6	23.8	19.1	14.2	7.7
		MCDB	8.2	10.5	14.4	18.4	22.2	24.8	29.8	30.1	26.8	21.7	17.1	10.8
	10%	WB	3.9	5.9	9.8	14.1	17.8	21.6	25.5	26.0	22.8	17.7	12.7	6.3
		MCDB	6.8	9.5	13.2	17.3	21.4	24.4	28.9	29.4	25.7	21.1	15.9	9.7
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	8.1	8.3	8.1	6.3	5.4	5.8	6.5	7.5	7.4	7.3
		MCDBR	8.2	9.2	9.6	10.6	10.8	8.7	7.7	7.5	7.7	8.2	8.5	8.8
		MCWBR	5.4	5.9	5.6	5.2	4.7	3.7	3.1	2.9	3.3	4.0	5.1	5.9
	5% WB	MCDBR	7.4	8.1	8.1	7.8	7.9	5.8	7.0	6.7	6.2	6.5	7.4	8.1
		MCWBR	5.8	6.5	6.3	5.6	4.8	3.5	3.3	3.0	3.5	4.4	5.7	6.3
Clear-Sky Solar Irradiance	taub	0.407	0.457	0.532	0.562	0.576	0.621	0.589	0.547	0.492	0.456	0.427	0.401	
	taud	2.078	1.982	1.819	1.800	1.820	1.790	1.927	2.035	2.135	2.114	2.107	2.109	
	Ebn at Noon	758	763	739	742	739	704	724	747	769	757	732	737	
	Edh at Noon	127	156	200	214	213	219	190	167	144	135	122	116	
All-Sky Solar Radiation	RadAvg	2.75	3.40	4.40	5.03	5.55	4.91	4.58	4.64	3.97	3.69	2.79	2.49	
	RadStd	0.21	0.34	0.39	0.39	0.58	0.43	0.67	0.58	0.42	0.34	0.32	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+64	N/A

Nomenclature: See separate page